Reconstructing Human Education in the Age of Artificial Intelligence

A Global Framework for Human-Centered Education in the Age of Artificial Intelligence

This document is an independent research outcome developed by the author, Xufen Tu, through long-term interdisciplinary study,

Release Date: 8 February 2026.

This document does not represent any government, institution, or policy position,

nor does it constitute a binding normative or enforceable instrument,

and is intended solely for research, education, and public reference.

Preamble

Artificial intelligence technologies are integrating into all levels of human society at an unprecedented pace. Activities such as information processing, resource allocation, risk assessment, and decision support are increasingly being carried out with the participation of algorithmic systems. From public governance to commercial operations, from educational environments to personal life, technical systems are gradually becoming embedded in critical links that influence real-world consequences.

This process has brought significant gains in efficiency, expansion of scale, and strengthened coordination capacity, while also triggering a series of deep structural changes. Decision processes are being reorganized, chains of responsibility are being redistributed, and the position and function of individuals within systems are shifting. Certain links that previously relied on human judgment and experience are being formalized, automated, and outsourced to technical systems for handling.

At this historical stage, societies across countries differ substantially in cultural traditions, institutional arrangements, and development paths, yet they jointly

face a fundamental question that cannot be avoided: when technical systems can provide recommendations, predictions, and execution plans in an increasing range of scenarios, how can humans maintain a clear role of judgment and a position as the responsible subject within those systems.

This issue does not arise from a denial of technological capability, but from concern for long-term social stability. Historical experience shows that when responsibility is no longer clear, judgment is no longer traceable, and agency is no longer explicit, systemic risk tends to accumulate. Technology itself does not automatically provide value direction or bear consequences; its use and embedded structures still require human societies to exercise judgment.

Therefore, in a highly automated and intelligent environment, education is assigned a new mission for the era. Education is not only about the transmission of knowledge and skills, but also about how humans understand their position within technical systems, how they make choices in complex, uncertain, and irreversible circumstances, and how they bear responsibility for those choices.

This framework aims to provide a long-term, stable, and globally adoptable educational reference structure for this challenge. Its goal is not to restrict technological development, nor does it attempt to provide a single unified model for different societies, but to ensure that the development of artificial intelligence can, within clear human boundaries and responsibility structures, mutually support and reinforce the continuous development of human judgment capability.

Executive Summary

This document proposes a framework for reconstructing human education in the age of artificial intelligence, responding to shared challenges faced by global society under rapid technological development. The framework is based on a fundamental position: artificial intelligence can significantly extend human capabilities, but it cannot replace human judgment.

Judgment is not merely a cognitive or computational activity, but a human act closely bound to responsibility, value trade-offs, and real-world consequences. Judgment occurs under conditions of incomplete information, complex contexts, and potentially irreversible outcomes. Its core feature is that a specific subject makes a choice and bears the consequences of the impacts that choice produces.

Artificial intelligence systems can play an important supportive role in judgment, including information integration, contextual analysis, and risk signaling. However, because technical systems do not bear consequences, they cannot be the final subject of judgment responsibility. This distinction is not a matter of higher or lower technical capability, but a fundamental difference in responsibility structure.

As artificial intelligence continues to evolve, the fundamental mission of education must adjust accordingly. Education can no longer be centered solely on knowledge transmission or skills training; it should focus more on the formation, maintenance, and inheritance of judgment capability. Through education, individuals should be able to understand when they can rely on technical support, when human judgment must intervene, and how to maintain clear responsibility boundaries in a technical environment.

This framework takes global applicability as its principle. It does not output specific institutional arrangements or ideological positions, but instead emphasizes a minimal level of human consensus. Regardless of cultural background or social system, when decisions have real-world consequences and affect others or society as a whole, human judgment cannot be fully outsourced.

By clarifying this consensus, this framework aims to provide long-term reference for national education systems, research institutions, and related practices, so that while artificial intelligence serves human society, human judgment capability, responsibility awareness, and civilizational confidence can be continuously consolidated.

Chapter 1: Structural Transformation in the Age of Artificial Intelligence

The development of artificial intelligence is changing the basic structure of how society operates. Decision processes are gradually being decomposed into three relatively independent links: data input, model analysis, and process execution. In this process, algorithmic systems begin to assume functions such as information screening, probabilistic assessment, and plan recommendation, and many judgment links that previously relied on human experience, intuition, and contextual understanding are being systematized.

This transformation is not a simple technological upgrade, but a reconfiguration at the level of social structure. As the degree of automation increases, the

functional boundaries of different roles within the decision chain are changing. The position of humans in systems is gradually shifting from judgment and bearing responsibility to use, supervision, and coordination. The increase in system efficiency makes this shift appear reasonable and attractive in the short term.

However, from a long-term perspective, this structural change also brings new risks. When judgment is gradually treated as an outsourceable technical function rather than a responsibility-bearing act that must be carried by a specific subject, the human role in critical decisions may be weakened unintentionally. System operation itself will not automatically preserve a place for human judgment unless that place is explicitly designed, educated, and maintained.

It is worth noting that this change is not caused by a single technological decision or policy choice, but is driven by multiple factors, including improved technical capabilities, organizational pursuit of efficiency, and complex systems' preference for consistency. In the absence of clear educational guidance, individuals often tend to trust system outputs rather than reflect on their own responsibility role within decision processes.

Therefore, the structural transformation in the age of artificial intelligence is not only a question of "whether humans still participate in decision-making," but rather "how humans participate, and at which links they bear judgment responsibility." The answer to this question will largely determine future social stability and the subject position of humans in technical environments.

Chapter 2: Shared Limitations Faced by Global Education Systems

Current education systems globally are primarily organized around knowledge transmission, skills training, and standardized assessment. This model played an important role in the industrial and informational stages by providing large-scale and replicable pathways for talent cultivation. However, under conditions of highly advanced artificial intelligence, the limitations of this educational structure are gradually becoming evident.

With the spread of information technology, knowledge and information themselves are no longer scarce. Artificial intelligence systems can rapidly acquire, integrate, and present large amounts of information, reducing the value of education aimed solely at memorization and reproduction. At the same

time, real-world situations requiring human judgment are becoming increasingly complex, often accompanied by uncertainty, value conflicts, and irreversible consequences.

In this context, the importance of judgment and responsibility is rising. Yet most education systems have not systematically cultivated individuals' abilities to form judgments in uncertain environments, understand potential consequences, and bear responsibility. Educational assessment still tends to reward standard answers, consistent performance, and short-term outcomes, while paying less attention to the formation and maturity of the judgment process itself.

This structural limitation is not a special case of any single country or culture, but a shared challenge for global education systems facing rapid technological change. Different societies vary in curricula and institutional arrangements, but broadly face the same issue: misalignment between educational objectives and real decision environments.

When education continues to reinforce the concept of a "correct solution" without sufficiently guiding individuals to understand that many real-world problems do not have a single optimal answer, individuals facing complex decisions often tend to rely on system recommendations rather than bear judgment responsibility. This tendency may reduce decision costs in the short term, but may weaken society's ability to identify responsible subjects in the long term.

Therefore, the challenge facing global education systems is not only about updating curricular content, but about whether educational objectives themselves can respond to the new requirements for judgment, responsibility, and human roles in the age of artificial intelligence.

Chapter 3: Technological Progress and the Repositioning of Human Roles

Artificial intelligence systems demonstrate advantages in speed, scale, and consistency that humans cannot match. Through large-scale data processing, model computation, and continuous optimization, technical systems can complete complex analyses in short periods and maintain stable performance in tasks with high repetition and consistency requirements. These capabilities have significantly improved efficiency in many fields, reduced costs, and expanded the possibilities of social operation.

However, technical advantage does not imply natural role replacement. Speed, scale, and consistency address the question of “how to complete a given task faster and more stably,” not “which tasks should be executed” or “which consequences should be accepted.” When decisions involve value trade-offs, risk bearing, or irreversible impacts, efficiency alone cannot provide a basis for judgment.

The unique human role is embodied precisely in this difference. Humans can make choices under conditions of incomplete information, unclear objectives, or conflicting values, and bear responsibility for the consequences those choices produce. Judgment does not arise from computational power advantages, but from understanding context, awareness of risk, and commitment to responsibility.

In an environment where artificial intelligence deeply participates in decision processes, if role boundaries are not clearly defined, technical systems are often misused as judgment authorities. Individuals may inadvertently cede judgment responsibility to system outputs, treating technical recommendations as neutral conclusions while ignoring the assumptions, constraints, and uncertainties embedded within them.

This role confusion does not originate from technical error, but from insufficient educational repositioning of human roles. When education fails to help individuals understand their judgment responsibility within technical systems, the human role is easily reduced to an “inefficient executor” or “final confirmer of the system.” Under such a cognitive structure, judgment is no longer seen as a responsibility act that must be carried, but as a burden that can be optimized, replaced, or avoided.

From the perspective of society as a whole, such change weakens the stability of responsibility structures. When judgment is no longer clearly attributed to specific subjects, society finds it difficult to identify decision sources, build long-term trust, and correct systemic deviations in time. Technical systems themselves cannot fill this gap, because they do not possess the capacity to bear consequences.

Therefore, the repositioning of human roles in the age of artificial intelligence does not require humans to compete with technology in capability, but requires humans to clarify their irreplaceable responsibilities within collaborative relationships. Education plays a key role in this process. Through education, individuals should be able to understand the advantages and boundaries of artificial intelligence, clarify their judgment position within decision chains, and

maintain sober awareness of choices and consequences under technical support.

The repositioning of human roles is the foundation of social stability and long-term trust in the age of artificial intelligence. Only when judgment responsibility is clearly retained in human hands can technological progress truly serve the overall well-being of human society.

Chapter 4: The Definition and Nature of Human Judgment

Human judgment differs from calculation or prediction. Calculation is a process of finding optimal or acceptable solutions within established rules, clear goals, and quantifiable constraints, through algorithms. Prediction is a probabilistic inference about future outcomes based on historical data and model assumptions. Both rely on the completeness of rules, definability of goals, and comparability of outcomes.

Judgment occurs under different conditions. When rules are not fully determined, goals are ambiguous, or value trade-offs themselves become the problem, calculation and prediction cannot independently complete decisions. In such contexts, judgment is not about finding an optimal solution, but about making a choice amid uncertainty and accepting the consequences that choice brings.

The first core feature of judgment is contextuality. Each judgment is embedded in specific times, environments, and networks of relationships. Even if surface conditions appear similar, constraints, risk distributions, and value considerations may change. This makes judgment impossible to fully abstract into a set of rules that can be repeatedly invoked.

The second feature of judgment is non-full reproducibility. Judgment is not determined solely by input information; its formation is influenced by multiple factors, including the information available at the time, the subject’s understanding, and risk awareness. Even if one reconstructs the judgment background afterward, it is impossible to fully restore the uncertainty and choice pressures faced at the time. This difference distinguishes judgment from technical processes that can be repeatedly verified and optimized.

The third feature of judgment is responsibility binding. Once judgment is made, it implies bearing responsibility for its consequences. This responsibility is not “right or wrong” in the sense of technical performance, but an undertaking

relationship in the social and ethical sense. The value of judgment lies precisely in this responsibility binding, not in whether outcomes match any predetermined standard.

It is precisely because of contextuality, non-full reproducibility, and responsibility binding that judgment cannot be fully automated. Artificial intelligence systems can provide important support in judgment processes, including information organization, option comparison, and risk signaling, but because they do not bear consequences, they cannot be the final subject of judgment responsibility.

This distinction does not negate technical capability, but clarifies different functional roles. Treating judgment as a fully replaceable technical function often stems from misunderstanding the nature of judgment rather than from reasonable expectations of technical potential.

In a context where artificial intelligence deeply participates in social operation, clarifying the definition and nature of human judgment has foundational significance for education. If education fails to help individuals understand the fundamental difference between judgment and calculation or prediction, individuals will struggle to recognize their irreplaceable roles in technical environments and maintain responsibility awareness in complex decisions.

Therefore, educationally elucidating human judgment as an independent, core, and non-outsourceable capability is not an evasion of technological progress, but a stable human reference framework for technological progress. Only within this framework can expanded artificial intelligence capabilities be coordinated with long-term social stability and trust structures.

Chapter 5: The Intrinsic Link Between Judgment and Responsibility

Judgment and responsibility are inseparable. When a choice may lead to real-world consequences, especially irreversible consequences, responsibility must be borne by a specific subject. This principle is not an ethical proposition, but a basic condition for social operation. The existence of responsibility enables society to identify decision sources, allocate consequence impacts, and make corrections and adjustments when necessary.

In complex systems, responsibility is not equivalent to the quality of outcomes. A judgment later proven non-optimal does not necessarily constitute irresponsibility; conversely, a decision process lacking a clear responsible

subject may weaken social stability in the long term even if it produces acceptable short-term outcomes. The core function of responsibility is to maintain a traceable relationship between decisions and consequences.

Artificial intelligence systems can provide analysis, recommendations, and predictions during judgment, but they do not bear consequences. No matter how precise system outputs are, the social impacts produced by operational results still must be borne by humans. This fact determines that artificial intelligence cannot be a responsible subject and cannot replace the final position of humans in the judgment chain.

In practice, responsibility ambiguity often stems not from technical failure, but from unconscious transfer of judgment responsibility. When system recommendations are treated as neutral conclusions or defaulted as “optimal choices,” humans may inadvertently relinquish judgment responsibility and attribute consequences to technology. This phenomenon is particularly common in highly complex and highly specialized environments.

Education plays a key role here. If education fails to help individuals understand the intrinsic link between judgment and responsibility, individuals often struggle to realize that they still bear judgment responsibility when facing technical systems. As a result, judgment is misunderstood as a burden that can be optimized, avoided, or outsourced rather than as a social role that must be carried.

Therefore, education should guide individuals to recognize that the value of judgment does not lie in arriving at a “correct answer,” but in clearly realizing the risks, constraints, and potential consequences borne when making choices. Bearing responsibility does not mean guaranteeing results, but maintaining sobriety and honesty about the choice process.

In the age of artificial intelligence, incorporating the intrinsic link between judgment and responsibility into educational goals helps maintain society’s ability to identify decision subjects. This identification ability is the foundation of public trust, institutional stability, and long-term technical usability.

Strengthening judgment responsibility awareness through education does not hinder technology application, but provides the necessary social framework for it. Only when responsible subjects remain clear can artificial intelligence support human judgment without becoming a tool for responsibility transfer.

Chapter 6: A Positive Definition of Human Boundaries

This framework does not treat human boundaries as restrictions on technological development, but as prerequisite conditions for effective collaboration between humans and artificial intelligence. Clear boundaries do not hinder technology application; they are the basis for ensuring that technology can play a positive role within social systems in a long-term and stable manner.

In environments where artificial intelligence participates in decision-making and execution, absence of boundaries often leads to role confusion. When technical systems are assigned functions beyond their capacity to bear responsibility, judgment and responsibility may be transferred unintentionally, weakening society's ability to identify decision subjects. By contrast, clear human boundaries help prevent technology from being misused as an authority source and provide stable structure for collaboration.

Artificial intelligence should be viewed as a supportive system, used to expand informational horizons, enhance analytical capabilities, and present possibility spaces. By integrating large amounts of data, simulating different contexts, and signaling potential risks, technical systems can significantly enhance human understanding of complex problems. However, such support does not equate to replacing judgment. Technical outputs themselves do not contain commitments regarding values, responsibility, or consequence bearing.

Humans, meanwhile, retain judgment authority and responsibility at critical nodes. These nodes typically involve value trade-offs, risk bearing, and irreversible impacts. When decision outcomes may affect others, society, or long-term structures, judgment must be made by specific human subjects who explicitly bear corresponding responsibility. This distinction is not based on relative capability, but on fundamental differences in responsibility attributes.

A positive definition of human boundaries helps form stable division of labor. In this relationship, artificial intelligence is encouraged to fully utilize its computational and analytical advantages, while humans focus on judgment, interpretation, and responsibility bearing. The two are not competitors, but functionally complementary.

From an educational perspective, clarifying human boundaries has foundational significance. If education emphasizes only technical usage skills without simultaneously explaining the principle of reserving judgment and

responsibility, individuals may mistakenly treat technical recommendations as final decision bases in practice. Conversely, by clearly expressing human boundaries through education, individuals can maintain judgment initiative under technical support and form healthy trust relationships in collaboration.

Such boundary definition also supports long-term social stability. When human judgment responsibility is explicitly retained, the introduction of technical systems will no longer weaken responsibility structures, but enhance overall decision capability. Society can thereby enjoy the efficiency and innovation brought by technological progress while maintaining clear recognition of consequence bearing and public trust.

Therefore, incorporating human boundaries as a positive principle into educational frameworks is not a defensive reaction against technological development, but a foundation for long-term coexistence between technology and human society. Only within clear boundaries can artificial intelligence continue as a tool serving humans, while human judgment capability is consolidated and developed within technical environments.

Chapter 7: The Core Mission of Human Education in the New Era

In the age of artificial intelligence, the core mission of education is no longer only to transmit knowledge or train skills, but to support the formation and maturity of judgment capability. As technical systems are widely applied in information processing and task execution, knowledge acquisition and skill operation are no longer the primary scarce human capabilities. Accordingly, the importance of judgment, responsibility, and contextual understanding is rising.

Education in this context bears the key task of repositioning human roles. Through education, individuals should be able to understand their position within technical systems, clarifying when to rely on technical support and when human judgment must intervene. This understanding is not an abstract concept, but the basis that shapes individuals’ behavior in real decisions.

Supporting the formation of judgment capability means education must focus on how individuals understand problems, identify risks, and make choices under uncertain conditions. This process is not aimed at providing standard answers, but at helping individuals develop awareness of the judgment process itself. Educational value thus shifts from result orientation to process orientation, from short-term performance to long-term maturity.

In environments where artificial intelligence participates in decision-making, individuals lacking clear cognition of judgment and responsibility often treat technical outputs as authoritative conclusions, weakening their own judgment roles. If education cannot respond to this change, individuals may gradually lose subjectivity within technical systems and begin to doubt their own value.

Therefore, human education in the new era should help individuals build stable judgment confidence. Such confidence does not come from comparing capabilities with technology, but from understanding one's irreplaceable roles. By clarifying the principle of reserving judgment and responsibility, education can help individuals maintain subjectivity in technical environments and form healthy trust structures in collaboration.

In addition, education should support individuals in understanding that judgment is not a mechanism guaranteeing correct outcomes, but an act of bearing uncertainty and consequences. With this understanding, individuals can remain rational and restrained when facing complex problems rather than relying on systems or authorities to avoid responsibility.

In sum, the core mission of human education in the new era is to support the formation of judgment capability, strengthen responsibility awareness, and maintain a clear position of humans as judgment subjects in societies where artificial intelligence is widely applied. This mission provides the basic direction for designing subsequent educational structures and implementation pathways.

Chapter 8: Rebuilding Human Confidence and Subjectivity

Human confidence does not come from comparisons with technological capability, but from confirmation of one's irreplaceable roles. Under rapid artificial intelligence development, if confidence is understood as a competition outcome in speed, accuracy, or processing scale, humans will inevitably be disadvantaged. This comparative path lacks realistic foundation and can weaken individuals' long-term recognition of their own value.

By contrast, stable and sustainable human confidence is built upon clear understanding of judgment and responsibility roles. When individuals recognize that their roles within technical systems are not being replaced but repositioned, the source of confidence shifts from capability confrontation to role confirmation. This confirmation does not depend on the level of technical proficiency, but on understanding one's judgment position.

Technological anxiety often stems not from technology itself, but from role uncertainty. When individuals cannot determine their responsibility boundaries within systems, they may project unease as fear of or resistance to technology. If education can clearly articulate the value and boundaries of human judgment, it can alleviate such anxiety and help individuals remain rational and stable when facing technical systems.

Rebuilding human confidence also means restoring subjectivity. Subjectivity is not total autonomy or rejection of collaboration, but the ability to make choices and bear consequences within collaborative relationships. The introduction of artificial intelligence, without educational guidance, may unintentionally weaken subjectivity, leading individuals to see themselves as replaceable components within systems. Conversely, by clarifying subject roles through education, individuals can maintain participation and responsibility under technical support.

In complex social systems, subjectivity is closely related to participation. When individuals understand that their judgment meaningfully affects system operation, they are more likely to sustain willingness to participate in public affairs, organizational decisions, and social issues. This participation is based not on control of outcomes, but on bearing judgment responsibility.

Therefore, clarifying the value of human judgment through education not only supports psychological stability at the individual level, but also supports trust and collaboration at the social level. When individuals generally possess clear cognition of their judgment roles, society as a whole can form stable responsibility structures and reduce overreliance on technical authority.

Rebuilding human confidence and subjectivity is not resistance to technological progress, but a prerequisite for technology to be long-term accepted and effectively utilized. Only when humans maintain confidence and subjectivity in technical environments can artificial intelligence continue as a supportive tool creating positive value for society.

Chapter 9: Cultivating Judgment Awareness

The formation of judgment capability does not begin with complex analysis or professional training, but with awareness of judgment itself. Education should first guide individuals to recognize when they are making judgments and when they are merely executing established processes. Awareness of the existence of choice is the starting point of judgment development.

In highly institutionalized and automated environments, many behaviors are embedded into processes. Individuals often complete tasks without realizing that choice spaces still exist. When judgment is concealed behind processes, individuals may mistakenly believe they are merely executors, ignoring the existence of judgment responsibility.

The introduction of artificial intelligence further amplifies this phenomenon. When system outputs are presented as recommendations, scores, or optimal plans, individuals lacking judgment awareness may treat them as conclusions beyond question. At such times, judgment has not disappeared, but has been silently completed and transferred, while responsibility remains unrecognized.

The primary educational task at this stage is not to teach how to make “better judgments,” but to help individuals become aware of when judgment is occurring. By distinguishing judgment from execution, individuals can gradually understand that in technical support environments, choice has not disappeared but appears in more hidden forms.

Cultivating judgment awareness helps individuals recognize that not all situations have standard answers. In many real-world problems, objectives may not be fully aligned, risk distributions may be difficult to quantify, and results may be uncertain. Awareness of these conditions helps individuals understand the necessity and complexity of judgment.

From a long-term perspective, societies lacking judgment awareness are more likely to attribute responsibility to systems, processes, or environments, weakening responsibility structures. Conversely, when individuals generally have judgment awareness, society can more clearly identify decision sources and maintain adaptability in complex environments.

Therefore, making the cultivation of judgment awareness a foundational educational link helps build a solid base for subsequent judgment capability development. This cultivation does not require individuals to always decide independently, but requires that within collaborative and supportive environments, individuals can always recognize the existence of their judgment roles.

By systematically cultivating judgment awareness through education, humans can maintain sober recognition of choice, responsibility, and consequences in societies where artificial intelligence is widely applied. This recognition is a prerequisite for sustained judgment capability and the foundation for maintaining human subjectivity.

Chapter 10: Judgment Boundaries and the Limits of Automation

Not all decisions are suitable for full automation. Artificial intelligence systems can significantly improve efficiency and consistency in tasks that are repeatable, quantifiable, and governed by clear rules. However, when decisions involve high risk, irreversible consequences, or value conflicts, there are clear limits to the applicability of automation.

The existence of judgment boundaries is not based on doubt about technical capability, but on distinctions in decision nature. When decision consequences may produce irreversible impacts on individual life, social structures, or long-term trust, reliance solely on technical system outputs cannot satisfy responsibility requirements. In such contexts, judgment is not merely a technical act, but a responsibility act.

High-risk situations are characterized by outcomes that, once occurring, are difficult to fully eliminate through later correction. Even if technical systems provide probabilistic assessments or risk alerts, their outputs still cannot bear the real-world consequences arising from choices. Therefore, in high-risk decisions, human judgment must exist as the final intervention point.

Irreversible situations further reinforce this requirement. When decision outcomes cannot be rolled back or compensated, the value of judgment lies not in predictive accuracy, but in conscious bearing of risk. Artificial intelligence can assist in identifying possibility spaces, but cannot substitute for human commitment regarding irreversible consequences.

Value-conflict situations constitute another important dimension of automation limits. In many real decisions, trade-offs among different objectives cannot be fully quantified. Values such as efficiency, fairness, safety, and dignity may conflict, and technical systems cannot determine which value should take priority. Here, judgment necessarily involves value choice, and value choice must be borne by human subjects.

Education’s role here is not to require individuals to reject automation, but to help them identify the scope and limits of automation applicability. Through education, individuals should understand when decisions can be handled by systems and when automated processes must be paused for human judgment intervention.

Without such boundary awareness, automation may be expanded unconditionally in practice, leading to hidden transfer of judgment

responsibility. Individuals may inadvertently outsource high-risk or value-laden decisions to systems, and when problematic outcomes occur, responsible subjects may be difficult to identify.

Therefore, explicitly incorporating judgment boundaries and automation limits into education supports the stability and explainability of decision systems. This clarity does not reduce technical efficiency, but provides a clear framework so automation can发挥优势 in appropriate contexts while avoiding roles beyond its responsibility-bearing capacity.

As artificial intelligence continues to evolve, specific technical forms may change, but the principles of judgment boundaries will remain valid over the long term. By cultivating understanding of these principles, education helps society maintain sober awareness of responsibility, risk, and values throughout technological development.

Chapter 11: The Formation and Expression of Judgment Structure

Judgment is not a random subjective reaction, but a structural process that can be described, reflected upon, and communicated. Although judgment occurs under uncertain conditions, its formation is not completely unintelligible or undiscussable. On the contrary, judgment is socially acceptable precisely because its process can be expressed and examined to some extent.

The structure of judgment usually includes understanding the context, identifying available pathways, awareness of risks and constraints, and anticipating possible consequences. These elements need not be quantified into unified standards, but can be articulated, recorded, and communicated among different subjects.

In the absence of judgment expression capability, judgment is easily misunderstood as arbitrary or emotional behavior. Such misunderstanding not only weakens the social legitimacy of judgment, but may push organizations and systems to rely on technical outputs as authority sources. If education cannot support expression of judgment structure, it may inadvertently reinforce reliance on automation.

By guiding individuals to learn how to express judgment reasons and constraint conditions, education helps transform judgment from an unspeakable personal experience into a socially understandable and discussable act. Such expression does not mean judgment must be proven “correct,” but means

others can understand the conditions and constraints under which judgment was made.

An important aspect of judgment expression is clearly distinguishing factual information from value orientation. Education should help individuals recognize which parts originate from information analysis and which involve value choice or risk bearing. This distinction reduces misunderstanding in collaboration environments and supports discussion among different subjects on the basis of respecting each other's judgments.

In addition, expressing judgment also helps form organizational and societal learning mechanisms. When judgment processes are recorded and communicated, society can accumulate experience and understanding without negating responsibility subjects. This accumulation differs from model training; it is experience inheritance based on responsible subjects.

In environments where artificial intelligence participates in decision-making, expressing judgment structure is especially important. Technical systems often present only output results, while judgment must maintain understandability through explaining process. By strengthening judgment expression capability, education helps individuals maintain control and explanation ability in decisions under technical support.

Therefore, incorporating the formation and expression of judgment structure into educational goals helps maintain decision explainability and social trust in uncertain environments. This capability ensures judgment neither degenerates into unquestionable authority nor is misunderstood as arbitrary subjectivity, but remains a human capability that is bearable, communicable, and reflectable.

Chapter 12: Judgment Records and Continuity

Recording judgment processes helps maintain traceability of responsibility. In complex systems, outcomes are often influenced by multiple factors, and recording outcomes alone cannot reflect the real conditions under which judgment occurred. By contrast, recording judgment conditions, information states, and constraint factors provides necessary context for understanding decisions.

The core purpose of judgment recording is not retrospective right-or-wrong evaluation, but maintaining continuity between judgment and context. By recording information available at the time, uncertainties faced, and alternative

paths considered, judgment can be placed within its real context, avoiding reduction of complex choices to single outcomes.

In the absence of judgment records, responsibility often becomes blurred over time. Later participants may evaluate judgments based solely on outcomes, ignoring constraints and risks at the time. Such outcome-oriented retrospection not only makes fair evaluation difficult, but may reduce individuals' willingness to bear judgment responsibility.

Education plays a foundational role here. Through education, individuals should be encouraged to record key elements of judgment processes, including information ranges relied upon, options excluded, and risk understandings at the time. Such records need not be exhaustive, but should aim to preserve judgment continuity.

Another important function of judgment records is supporting reflection and learning. When judgment processes are recorded, individuals and organizations can review and adjust judgments without negating responsibility subjects. This reflection differs from model calibration; it is experience accumulation grounded in responsibility bearing.

In environments where artificial intelligence participates in decisions, judgment records are particularly important. Technical systems often preserve inputs and outputs but ignore human choices within them. Recording judgment processes clearly distinguishes technical support from human judgment, preventing responsibility from being silently transferred to systems.

From a longer-term perspective, judgment records form part of social memory. Societies can understand choices made in different periods and conditions, maintaining continuity amid changing environments. Such continuity is an important basis for civilization to self-correct and stably evolve.

Therefore, making judgment records and continuity part of educational goals is not adding burden, but providing support for long-term development of judgment capability. By cultivating understanding of judgment recording through education, societies can maintain clear responsibility, transmissible experience, and stable continuation of judgment capability during ongoing technological evolution.

Chapter 13: Distinguishing Judgment Responsibility and Execution Responsibility

In complex systems, judgment responsibility and execution responsibility are often separated. With expanding organizational scale, high professionalization of processes, and the introduction of artificial intelligence systems, decisions and execution are often completed by different subjects at different times. Such separation is not inherently problematic; it is a necessary condition for complex social operation.

Judgment responsibility refers to responsibility for choice direction and consequences when facing uncertainty, risk, and value trade-offs. Execution responsibility refers to responsibility for completing tasks according to rules and processes within established judgment and instruction frameworks. The two are different in nature but are easily conflated in practice.

When judgment responsibility and execution responsibility are not clearly distinguished, responsibility tends to be diluted within systems. Executors may be forced to bear consequences for judgments they did not make, while judgment makers may disappear within execution complexity. Such conflation harms individual fairness and weakens organizations' and societies' ability to identify responsible subjects.

Participation of artificial intelligence systems makes this problem more prominent. Technical systems often participate both in judgment support and execution control, and their outputs may directly trigger actions. Without clear responsibility distinction, system recommendations, organizational instructions, and personal judgments can become mixed, leading to blurred responsibility boundaries.

Education plays a key role here. Through education, individuals should understand the existence of different responsibility levels and clarify the type of responsibility they bear. This understanding helps individuals neither evade judgment responsibility nor be improperly required to bear responsibilities beyond their roles.

Respecting the distinction between judgment responsibility and execution responsibility also supports stability of system operation. When judgment makers clearly bear directional responsibility and executors focus on execution quality, systems become more transparent and credible. By contrast, responsibility conflation often leads to defensive behavior, blame shifting, and decision delays.

From an educational perspective, this distinction does not require individuals to always occupy a single role, but requires that individuals can recognize the

type of responsibility they bear in different contexts. By strengthening this recognition ability, education helps individuals maintain clear responsibility awareness within complex systems.

In the future of widespread artificial intelligence application, the distinction between judgment responsibility and execution responsibility will become even more important. Technical systems can support execution efficiency, but cannot bear judgment responsibility. Only when education systematically conveys this principle can societies maintain clarity and stability of responsibility structures in highly automated environments.

Therefore, explicitly incorporating the distinction between judgment responsibility and execution responsibility into educational frameworks helps prevent responsibility from being silently diluted within systems. This distinction not only protects individuals but provides organizations and societies with a long-term trustworthy responsibility foundation.

Chapter 14: Artificial Intelligence as a Supportive System

The legitimate role of artificial intelligence in education is information integration, scenario simulation, and cognitive extension. By processing large amounts of data, presenting multiple possibilities, and simulating different contexts, artificial intelligence can significantly enhance learners' ability to understand complex problems. However, the essence of this capability is to support judgment, not replace judgment.

As a supportive system, the function of artificial intelligence is to expand the range of information humans can see, not to reduce choice space. Technical systems can help learners identify potential variables, compare different options, and understand risk distributions, but their outputs do not constitute commitments to choice direction. Final formation of judgment still must be completed by human subjects.

In educational environments, if artificial intelligence is implicitly granted authority status, learning processes may shift unintentionally. Learners may treat system outputs as correct answers or optimal paths, ignoring responsibility awareness and contextual understanding required by judgment. This shift is not technical failure, but a structural issue resulting from unclear role positioning.

Therefore, clearly defining artificial intelligence as a supportive system helps preserve the core educational objective. Technology is used to extend cognitive boundaries, not to compress judgment space. Under this positioning, learners can maintain respect for problem complexity under technical assistance and understand that judgment does not disappear because information becomes abundant.

Artificial intelligence as a supportive system can also provide new modes of expression and exploration for education. Through scenario simulation, learners can experience consequences of different choices in low-risk environments, deepening understanding of judgment conditions and constraints. Such simulation does not replace real judgment, but provides safe practice space for judgment capability formation.

At the same time, education must clarify functional boundaries of artificial intelligence within learning processes. Systems do not bear consequences of learners' judgments, nor provide final bases for value trade-offs. This distinction prevents responsibility transfer within learning and maintains education's respect for subjectivity.

From a long-term perspective, positioning artificial intelligence as a supportive system promotes sustainable application in education. When technology is understood as an assisting tool rather than authority source, its introduction will not weaken human judgment capability but can strengthen learners' understanding and control of judgment processes.

Therefore, clarifying the supportive role of artificial intelligence in education is an important condition for maintaining stable educational direction under rapid technological development. With this positioning, artificial intelligence can continue to serve education while the core mission of education—supporting development of human judgment capability—is fully preserved.

Chapter 15: Functional Boundaries of Artificial Intelligence

Education should clarify that artificial intelligence does not bear final value judgment and responsibility. This boundary is not a restriction on technical capability, but a clear definition of technical roles. Only when functional boundaries are clear can artificial intelligence be stably and credibly incorporated into education and social systems.

Artificial intelligence systems can provide analysis, recommendations, and predictions in many contexts, but these outputs are always built on preset objectives, training data, and model assumptions. Systems do not possess the ability to adjudicate value conflicts by themselves, nor do they bear real-world consequences produced by choices. Therefore, their outputs cannot be treated as substitutes for final judgments.

When this boundary is not clearly expressed, technology can be misused as an authority source. Learners or decision participants may treat system conclusions as neutral, objective results requiring no further responsibility, inadvertently abandoning judgment roles. This phenomenon does not arise from design defects, but from absence of boundary education.

Clarifying functional boundaries helps prevent hidden responsibility transfer. When systems are correctly understood as supportive tools, humans can maintain sober awareness of choice directions and consequences while using technical recommendations. Judgment responsibility is thereby clearly retained by human subjects rather than dispersed or blurred.

In educational environments, this boundary is particularly important. Education shapes not only skills but role cognition. If learning does not clarify that artificial intelligence does not bear final value judgment, learners may later treat technology as decision substitutes rather than assisting tools. This can produce long-term impacts on social responsibility structures.

By systematically conveying functional boundaries through education, individuals can gradually form rational attitudes toward technical outputs. This attitude is neither blind reliance nor rejection, but cautious application based on understanding limitations.

From a longer-term perspective, boundary clarity supports broad adoption of artificial intelligence across different societies, cultures, and institutional environments. Boundary principles do not depend on specific technical forms, so they remain effective across technological evolution. Such stability provides long-term reliable frameworks for education and society.

Therefore, incorporating functional boundaries into educational frameworks is an important condition for preventing technology from being misused as authority and for maintaining human judgment and responsibility structures. Under this principle, artificial intelligence can continue to provide support without eroding fundamental societal understanding of judgment subjects.

Chapter 16: Long-Term Transformation of Assessment Methods

What corresponds to judgment-oriented education is a reconstruction of assessment methods. In environments where artificial intelligence can quickly provide answers, plans, and optimal paths, relying solely on outcome correctness as the primary measure of educational achievement no longer reflects individuals' real abilities in actual decision contexts.

Traditional assessment systems have advantages in standardization, comparability, and scalability, but their focus on outcomes and efficiency often overlooks judgment processes themselves. When assessment only examines whether expected answers are produced, learners may reduce learning objectives to outcome matching rather than contextual understanding and responsibility awareness formation.

Within judgment-oriented education frameworks, assessment focus must undergo long-term transformation. Educational achievement should emphasize judgment maturity, stability, and responsibility awareness rather than correctness of single conclusions. Judgment maturity is reflected in individuals' ability to recognize uncertainty, understand constraints, and make reasonable choices in complex contexts. Stability is reflected in maintaining consistent judgment principles across conditions rather than losing direction entirely with environmental changes.

Responsibility awareness is indispensable in judgment assessment. Assessment should not require individuals to guarantee successful outcomes, but should examine whether individuals clearly recognize risks and consequences borne and can provide honest accounts of judgment processes. Such awareness is the basis for judgment to be socially trusted.

Transformation of assessment does not negate the importance of knowledge and skills. Knowledge and skills remain foundations of judgment, but their value lies in supporting judgment rather than replacing it. Assessment systems must reflect this relationship rather than conflating knowledge mastery with judgment capability.

In practice, judgment-oriented assessment emphasizes process explainability and continuity. By focusing on how individuals form judgments, express constraints, and reflect afterward, assessment can more accurately reflect the developmental state of judgment capability. This approach helps avoid overly simplified evaluation based on single outcomes.

From a long-term social perspective, transformed assessment helps form more stable talent structures. When education rewards judgment maturity and responsibility awareness, society can more readily cultivate individuals capable of bearing responsibility in uncertain environments. Such capability will become an important component of social resilience in highly automated futures.

Therefore, gradually shifting assessment from outcome orientation to judgment maturity is not a short-term reform measure, but a structural adjustment for the future. By systematically advancing this transformation through education, societies can maintain long-term emphasis on judgment, responsibility, and subjectivity amid ongoing technological evolution.

Chapter 17: Education as an Investment in Social Resilience

Cultivating judgment capability is a long-term investment whose returns do not always appear immediately but profoundly shape social stability and adaptability in uncertain environments. Like infrastructure development, the value of judgment education is often most evident during periods of pressure, crisis, and transition.

Social resilience refers to society's ability to maintain basic functions and gradually recover when facing sudden events, systemic risks, or long-term uncertainty. This ability depends not only on technical resources and institutional design but also on the capacity of individuals and organizations to make judgments and bear responsibility at critical times.

In highly automated societies, technical systems can significantly improve response speed and execution efficiency, but their stability still depends on human judgment in abnormal conditions. When model assumptions fail, data conditions change, or objectives conflict, systems cannot decide directions by themselves. At such moments, judgment capability becomes key to preventing system disorder.

Education as an investment in social resilience functions at multiple levels. First, by cultivating judgment awareness and responsibility awareness, education enhances society's ability to recognize risks. Individuals no longer rely solely on existing processes but can proactively identify potential problems under changing conditions. Second, judgment education helps maintain social

trust. When decision processes are clear and responsible subjects explicit, the public more readily accepts uncertain outcomes, reducing systemic conflict.

From a long-term perspective, cultivating judgment capability also helps reduce recovery costs during crises. When individuals broadly have the ability to bear judgment responsibility, society can adjust strategies and correct directions more quickly without relying entirely on centralized commands or temporary authorities. This distributed judgment capability is an important source of resilience in modern complex societies.

However, compared with short-term, quantifiable technical investments, returns on judgment education have longer cycles and are difficult to measure with single metrics. This feature can lead it to be underestimated in policy and resource allocation. If education systems focus only on short-term performance and neglect long-term accumulation of judgment capability, they may weaken deep stability while improving surface efficiency.

Therefore, treating education as an investment in social resilience means providing sustained support for cultivating judgment capability in resource allocation and institutional design. Such support does not come at the expense of technological development but ensures synchronized development of technical capability and human judgment capability.

In the future of continued artificial intelligence evolution, societies will inevitably face more complex and unpredictable contexts. The widespread existence of judgment capability will determine whether societies can maintain direction and cohesion in such contexts. By systematically investing in judgment capability through education, human societies can maintain long-term stability and trust foundations amid rapid technological change.

Chapter 18: Cross-Cultural Applicability and Minimal Consensus

This framework does not output a unified value system, nor does it attempt to form standardized templates across culture, institutions, or development paths. Different societies have differences in historical background, social structure, and value orientation, which constitute the foundation of civilizational diversity. This framework respects and preserves such diversity.

At the same time, under conditions where artificial intelligence widely participates in social operation, different societies are facing certain structural problems in common. These problems arise not from cultural differences but

from technology reshaping decision structures, responsibility distribution, and subject roles. Therefore, this framework emphasizes minimal consensus rather than comprehensive unified norms.

This minimal consensus can be summarized as: when decisions have real-world consequences and affect individuals, groups, or society as a whole, judgment and responsibility cannot be fully automated. This principle does not rely on any specific moral system, political regime, or religious tradition, but on basic understanding of responsibility-bearing capacity.

In different cultures, the specific forms of expressing judgment and responsibility may vary. Some societies emphasize collective decision-making, others emphasize individual bearing; some institutions define responsibility through formal procedures, others through social networks. However, regardless of form, the requirement that responsibility must be attributable to specific subjects always exists.

This framework seeks cross-cultural applicability precisely at this point. It does not prescribe who should make judgments, but emphasizes that judgments must be made by subjects capable of bearing consequences. This distinction allows the framework to be adopted across different institutional environments without replacing existing value structures.

Artificial intelligence development is global, crossing borders. Without minimal consensus, different societies may form conflicting responsibility understandings in technical applications, increasing cross-system risks. By clarifying in education the principle that judgment and responsibility cannot be fully outsourced, societies can form communicable responsibility language globally.

Education has unique advantages in achieving cross-cultural applicability. Education does not directly intervene in institutional operation but shapes understanding and role cognition, providing shared cognitive foundations across societies. Conveying minimal consensus through education rather than specific models respects differences while maintaining basic stability of responsibility structures.

Therefore, by emphasizing minimal consensus rather than unified values, this framework provides flexible adoption space for different cultures and institutions. Such design allows it to remain applicable amid social change and technological evolution, supporting long-term coexistence between artificial intelligence and human judgment globally.

Chapter 19: Application Across Different Social Stages

Regardless of social development stage, judgment-oriented education has practical significance. Childhood education, adult re-education, and professional and public-domain education differ in objectives and forms, but all face the same structural issue: in environments where technology deeply participates in decisions, how to maintain continuous development of judgment capability and responsibility awareness.

In child and adolescent education, judgment orientation does not mean prematurely introducing complex decision models, but helping learners become aware of the existence of choice. By understanding that problems may have multiple paths and outcomes are not fully predictable, individuals can gradually form basic cognition of judgment. The focus at this stage is cultivating judgment awareness and expression capability, rather than evaluating outcome quality.

In adult re-education and career development stages, the significance of judgment orientation is more direct. As artificial intelligence tools enter work contexts widely, individuals need to re-understand their roles under technical support. Education at this stage should support individuals in identifying boundaries between judgment and execution, understanding the applicability of technical recommendations, and bearing judgment responsibility in complex contexts. Such capability is important for career stability and organizational trust.

In professional and public domains, judgment-oriented education relates to management of systemic risks. Decisions in medicine, engineering, finance, public governance, and other fields often involve high risk and irreversible consequences. In these fields, education does not replace professional norms but strengthens practitioners' understanding of judgment responsibility so technical support can function within clear human boundaries.

Societies at different development stages differ in resources, institutional maturity, and technical foundations. However, judgment-oriented education does not depend on high-cost facilities or specific technical environments. Its core lies in role cognition and responsibility understanding, so it can be implemented in different forms under various conditions. This flexibility enables the framework to be adopted across stages without imposing additional burdens.

In resource-limited environments, judgment-oriented education helps enhance system adaptability. When technical resources are insufficient or conditions unstable, judgment capability can compensate for lack of processes and tools. In highly technical environments, judgment-oriented education helps prevent overreliance on systems and maintain clarity of responsibility structures.

Therefore, Chapter 19 emphasizes not uniform implementation paths but shared directions. Regardless of development stage, once technology begins to influence decision structures, judgment-oriented education has practical significance. By continuously strengthening judgment and responsibility awareness across stages, societies can maintain continuity and stability during technological evolution.

Chapter 20: Educational Responsibility for the Next One Hundred Years

Technology will continue to evolve, and its forms, capabilities, and application scenarios will keep changing. Specific implementations of artificial intelligence may undergo multiple iterations, but their impacts on decision structures and social operation will persist long-term. Throughout this ongoing change, civilizational stability depends not on the strength of technology itself, but on whether humans can maintain judgment and responsibility in technical environments.

Historical experience shows that civilizations do not become stable automatically through tool progress. Rather, when tool capabilities expand rapidly while responsibility structures do not adjust accordingly, societies often face imbalance risks. The challenge raised by the age of artificial intelligence is precisely this structural issue under new technical conditions.

Education in this context bears intergenerational responsibility. Unlike technology, the impact of education is not measured by versions or cycles, but shapes understanding, role cognition, and responsibility awareness through long-term accumulation. Judgment capability formed through education will influence how future societies respond under uncertainty.

For the next one hundred years, education must bear not only the task of adapting to change, but also the task of maintaining continuity. Continuity does not mean staying unchanged, but preserving basic principles amid change so societies can reposition themselves under new conditions. The core of this

principle is that judgment and responsibility must always be retained by subjects capable of bearing consequences.

Artificial intelligence development provides unprecedented tools and possibilities for human society. But these possibilities can only be transformed into long-term value within clear responsibility structures. By continuously strengthening judgment capability and responsibility awareness, education ensures that technological progress does not erode the human subject position in decisions.

This responsibility does not require education to predict specific future technical forms, nor to provide certain answers for all problems. Rather, the long-term mission of education is to enable individuals and societies to make judgments under uncertainty and bear consequences for those judgments. This capability is not replaceable by any technology.

Therefore, educational responsibility for the next one hundred years is not to propose new authorities or norms, but to safeguard a basic fact: when choices have real-world consequences, humans must retain judgment and responsibility. This fact remains valid across historical stages, cultures, and technical conditions.

Conclusion

The development of artificial intelligence is an important achievement of human civilization. Its true value lies not in whether it can replace humans, but in whether it can support humans to make more responsible choices in a complex and uncertain world.

As technical capabilities continue to expand, the problems societies face do not disappear; instead, they become more hidden in form. Choices still exist, consequences remain real, and responsibility still must be borne by subjects capable of carrying its impacts. In this context, technological progress cannot automatically guarantee civilizational stability.

Education’s mission is to act precisely at this critical point. Education is not for predicting future technical forms, but for ensuring that regardless of how technology evolves, humans can understand their roles in decisions and continuously maintain judgment and responsibility awareness.

When judgment capability is cultivated systematically and responsibility is not obscured by technology, human society can maintain clear direction and trust

foundations in highly automated environments. The value of this capability is not determined by a single generation but is reflected in continuity across time.

When everything can be automated, humans must still be educated to judge.

Appendix A

Institutional Background and Framework Stewardship Statement

As an educational framework study oriented to the age of artificial intelligence, the purpose of this document is to propose long-term applicable educational reference principles rather than to formulate specific policies, curriculum standards, or implementation plans. The viewpoints and framework described herein aim to provide open theoretical and structural reference for different societies, education systems, and research institutions.

This framework is supported and maintained, as one of the initiating and hosting institutions for research and text organization, by **Oxygen Magic Academy (氧气魔法学院)**. As an interdisciplinary education and research platform, the Academy has long focused on the evolution of human judgment capability, responsibility structures, and educational models under conditions of technological development.

The role of Oxygen Magic Academy is not that of an authoritative issuing institution, nor does it hold exclusive interpretive authority over this framework. The Academy does not aim to output a unified educational model or value system, but is committed to providing a sustainable and revisable reference-text environment for relevant research, educational practice, and public discussion.

The content of this framework does not constitute any form of instructional directive, career commitment, or technical specification, nor does it guarantee specific implementation outcomes. It is positioned as long-term research and educational reference material that may be independently interpreted and adopted across different cultural, institutional, and educational contexts.

In the future, any citation, discussion, or extension of this framework should preserve its non-exclusive and open characteristics and respect its core principle: when decisions have real-world consequences, human judgment and responsibility cannot be fully automated.

Use Statement

This appendix serves only to describe the research and stewardship background of this framework and does not constitute any authorization, endorsement, or affiliation statement regarding any individual, organization, or institution.

Appendix B

Key Terms and Concept Notes

This appendix explains key terms used in this framework to ensure consistency of understanding across different cultural, institutional, and educational contexts. This appendix does not constitute formal definitions at legal, medical, technical, or policy levels; it is provided solely to support accurate interpretation of this text.

Judgment

Refers to a choice act made by a specific subject under conditions of incomplete information, uncertain objectives, or conflicting values. The core feature of judgment is responsibility binding: once judgment is made, the subject must bear its real-world consequences.

Responsibility

Refers to the bearing relationship for impacts produced by judgment or choices. Responsibility is not equivalent to success or failure of outcomes, but lies in maintaining traceability and bearing awareness regarding the choice process and its consequences.

Automation

Refers to processes executed or controlled by technical systems under established rules, objectives, and constraints. Automation is suitable for repeatable and quantifiable tasks, but does not possess the capacity to bear judgment responsibility.

Supportive System

Refers to technical systems used to extend capabilities for information acquisition, analysis, and understanding. Supportive systems may participate in judgment processes, but do not constitute judgment subjects and do not bear final responsibility.

Judgment Boundary

Refers to the structural position within decision processes where human judgment must intervene and retain responsibility, commonly occurring in high-risk, irreversible, or value trade-off contexts.

Appendix C

Scope, Limitations, and Non-Applicability Statement

This framework is a reference document for education and research. Its scope and limitations are as follows.

This framework does not constitute any form of policy directive, legal opinion, medical advice, or vocational training standard. Its content is provided solely as long-term educational principles and structural reference, and makes no guarantees regarding specific implementation methods, teaching outcomes, or social effects.

This framework does not replace existing education systems, professional norms, or industry standards. Its purpose is to supplement interpretive reference to support discussion and research on human judgment and responsibility in the age of artificial intelligence.

This framework does not make any compliance, effectiveness, or safety judgments regarding any specific technical systems, algorithmic models, or artificial intelligence applications. Responsibility for all technical applications remains with relevant implementing subjects under their respective institutional and legal environments.

This framework is not applicable as a design specification or execution basis for automated decision systems. Its content must not be used to justify any form of fully automated judgment, responsibility transfer, or system authority elevation.

Any citation, adaptation, or extension of this framework should clearly state its nature as a referential and non-binding text and preserve its core position that human judgment and responsibility cannot be fully automated.

Appendix D

Notes on Educational Outcomes and Real-World Social Linkage

This framework does not take specific occupations, positions, or employment structures as its target, nor does it predict the specific form of future labor markets. Its focus is on cultivating judgment capability and responsibility awareness that maintain stable value amid changing occupational structures.

Capabilities formed through judgment-oriented education have high transferability. These capabilities do not depend on any single industry or technical pathway and can continue to function across different social roles and work environments.

In practical application, judgment capability and responsibility awareness may be reflected across multiple domains, including but not limited to professional decision-making, organizational collaboration, public affairs participation, and complex systems management. This framework does not prioritize these domains, nor does it constrain specific forms.

By emphasizing capability transfer rather than position matching, this framework aims to support individuals in maintaining adaptability amid changes in technology and occupational structures. Educational outcomes thus are not bound to short-term employment demand but can serve long-term social stability and individual development.

This appendix is provided solely to explain the linkage logic between this framework and real-world society and does not constitute any employment commitment, career guidance, or capability certification standard.

Appendix E

Document Status, Versioning, and Revision Principles

As an educational framework study oriented to the age of artificial intelligence, this document is a **long-term reference text**, not a one-time policy document or execution standard. This appendix describes the document's status, version attributes, and possible future revision principles.

Document Status

This framework is a research and educational reference document. It aims to provide a cross-time-scale applicable principles-based structure to support continuous discussion on human judgment, responsibility, and the role of education. This document does not constitute a final conclusion, nor does it aim to exhaust all related topics.

This document is not a mandatory basis for any institution, organization, or individual. Its content may be read, cited, or discussed independently, but it does not carry binding or enforceable effect.

Versioning

The current version of this document is **Version 1.0**.

The version number serves only to mark publication sequence over time and does not represent the end state of theoretical maturity.

If revised versions are published in the future, their purpose is to improve clarity of expression, structural consistency, or applicability explanation, rather than to frequently chase short-term technological changes.

Revision Principles

Any revision to this framework should follow these basic principles:

1. Revisions should not change the core position of this framework, namely:

   When decisions have real-world consequences, human judgment and responsibility cannot be fully automated.

2. Revisions should not introduce dependence on specific technical routes, algorithmic models, or short-term application scenarios.
3. Revisions should aim to enhance cross-cultural understanding and long-term applicability, rather than adapt to specific institutional or policy environments.
4. Revisions should preserve the text's non-exclusive and open characteristics and should not establish a single authoritative interpretive source.

Use and Reference

This document may be used in research, educational discussion, and public communication contexts.

Any citation, summary, or redistribution should respect its positioning as a principles-based framework and must not present it as a policy command, legal

opinion, or technical specification.

The presence of this appendix does not limit academic discussion or critique of this document; it exists to ensure structural integrity and semantic stability throughout long-term dissemination.